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Exp 13.1:

Algorithm:

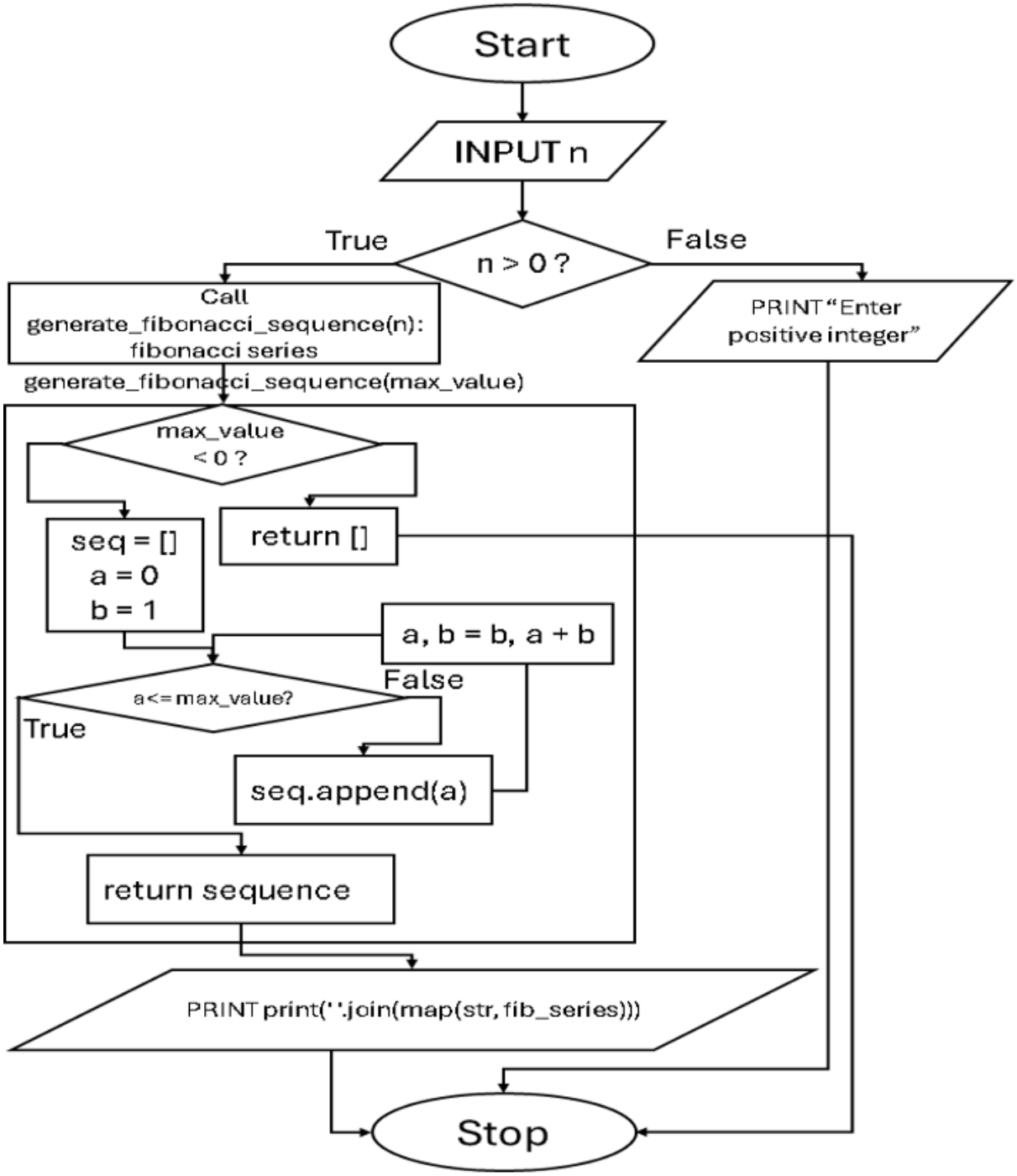
Step 1: Start.

Step 2: Read an integer input from the user and store it in a variable (let's call it max_value).

Step 3: Check if max_value is less than or equal to 0.

- If **True**: Print "Please enter a positive integer" and jump to Step 8.
- If **False**: Continue to Step 4.
- **Step 4:** Initialize two variables to represent the first two Fibonacci numbers: a = 0 and b = 1.
- **Step 5:** Create an empty list called sequence to store the numbers.
- **Step 6:** Loop while a is less than or equal to max_value:
- Add the current value of a to the sequence list.
- Calculate the next term by adding a and b together.
- Update the variables: set a to the value of b, and set b to the newly calculated next term.
- **Step 7:** Print the sequence as space-separated numbers.
  Step 8: Stop.

Flowchart:


Start
INPUT n
True
n > 0 ?
False
Call
generate_fibonacci_sequence(n):
fibonacci series
PRINT "Enter
positive integer"
generate_fibonacci_sequence(max_value)
max_value
< 0 ?
seq = []
a = 0
b = 1
return []
a, b = b, a + b
False
a<= max_value?
True
seq.append(a)
return sequence
PRINT print(' '.join(map(str, fib_series)))
Stop

Code:

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13.1.1. Fibonacci Sequence using Modules 29:31

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Module Requirements

- Create a module **fibonacci_module.py** with a function **generate_fibonacci_sequence(max_value)** that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create **main.py** that performs the following:

- Import the **fibonacci_module.**
- Read an integer n from user input.
- Validate if n is positive.
- If valid, call the module function and display the Fibonacci sequence.
- If invalid, display an error message.

Input Format:

- Single integer n representing the maximum value for Fibonacci sequence.

Output Format:

- If $n > 0$, print the Fibonacci sequence as space-separated numbers.
- If $n \leq 0$, print: "Please enter a positive integer".

Note:

- Two files are provided: **fibonacci_module.py** and **main.py**. The **main.py** file handles input and output. You need to complete both files based on the question requirements.

Explorer | main.py | fibonacci... | Submit | Debugger

```python
def generate_fibonacci_sequence(max_value):
    sequence = []

    a, b = 0, 1

    while a <= max_value:
        sequence.append(a)
        a, b = b, a + b

    return sequence
```

Average time **0.003 s** 2.75 ms | Maximum time **0.006 s** 6.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 6 ms Debug

Expected output	Actual output
10	10
0 1 1 2 3 5 8	0 1 1 2 3 5 8

Test case 2 3 ms

Test case 3 3 ms

Terminal | Test cases